

3 Safety

3.1 General

3.1.1 Disclaimer

The robot system described in this document is a robotic component for integration into a medical device (hereinafter called "LBR Med"):

The LBR Med includes:

- Robot
- Robot controller
- Teach pendant
- Connecting cables
- Software
- Options, accessories

The LBR Med is built using state-of-the-art technology and in accordance with the recognized safety rules. Nevertheless, misuse may constitute a risk to life and limb or cause damage to the LBR Med and to other material property.

The LBR Med may only be used in perfect technical condition in accordance with its intended use and only by safety-conscious persons who are fully aware of the risks involved in its operation. Use is subject to compliance with this document and with the documentation provided with the robot upon delivery (Important information on LBR Med). Any functional disorders, especially those affecting safety, must be rectified immediately.

The LBR Med meets the requirements of the standards named in the CB Test Certificate. All requirements deviating from the CB Test Certificate are not met.

All uses deviating from the CB Test Certificate have not been considered, examined and tested. The medical product manufacturer must evaluate these uses in its risk management process and take the necessary measures for the medical product / overall system.

Examples of such uses include:

- Integration into a portable or hand-held medical device
- Integration of the LBR Med into medical devices which are intended for sale to the general public
- Operation in domestic environments
- Operation in the vicinity of emergency response operations
- Operation in the vicinity of strong magnetic fields
- Operation in combination with the use of defibrillators (the robot is not defibrillator-safe)
- Unsupervised operation of the LBR Med or the medical device (e.g. by the medical device manufacturer or end user)
- Operation of the medical device by non-medical personnel, patients or third parties.
- Operation of the robot as an applied part
- Connection of type BF or CF applied parts according to IEC 60601-1 directly to the robot without suitable intermediate insulation
- Power supply to an applied part via the data cables of the integrated energy supply system (e.g. Power over Ethernet)
- Operation by persons without the relevant training

Safety information

Information about safety may not be construed against KUKA Deutschland GmbH. Even if all safety instructions are followed, this is not a guarantee that the LBR Med will not cause personal injuries or material damage.

No modifications may be carried out on the LBR Med without the authorization of KUKA Deutschland GmbH. Additional components (tools, software, etc.) not supplied by KUKA Deutschland GmbH may be integrated into the LBR Med. The user is liable for any damage these components may cause to the LBR Med or to other material property.

In addition to the Safety chapter, this document contains further safety instructions. These must also be observed.

3.1.1.1 CB Test Certificate and CB Test Report

The LBR Med is tested according to the CB Scheme and delivered with a CB Test Certificate.

Through the CB Test Certificate, it is declared that the LBR Med has been produced in accordance with the standards named in the CB Test Certificate.

Only the software components named in the CB Test Report in the specified versions are compatible with one another and may be jointly installed. If software components deviating from the CB Test Report are installed, the CB Test Report is no longer valid.

3.1.1.2 EC declaration of conformity and declaration of incorporation

The LBR Med constitutes partly completed machinery as defined by the EC Machinery Directive. The LBR Med supplied may only be put into operation if the following preconditions are met:

- The LBR Med is integrated into a medical device.
- The medical device complies with Medical Device Directive/Regulation. This has been confirmed by means of a conformity assessment procedure.

EC declaration of conformity

The system integrator must issue an EC declaration of conformity in accordance with the EU Directive/Regulation for the medical device. The EC declaration of conformity forms the basis for the CE mark for the medical device. The LBR Med must always be operated in accordance with the applicable national laws, regulations and standards.

The robot controller has a CE mark in accordance with the EMC Directive and the Low Voltage Directive.

Declaration of incorporation

The partly completed machinery is supplied with a declaration of incorporation in accordance with Annex II B of the Machinery Directive 2006/42/EC. The assembly instructions and a list of essential requirements complied with in accordance with Annex I are integral parts of this declaration of incorporation.

The declaration of incorporation declares that the start-up of the partly completed machinery is not allowed until the partly completed machinery has been incorporated into a medical device, or has been assembled with other parts to form a medical device, and this medical device complies with the terms of the Medical Device Directive/Regulation, and the EC declaration of conformity is present.

3.1.2 Terms in the “Safety” chapter

Term	Description
Axis range	Range within which the axis may move The axis range must be defined for each axis.
Stopping distance	Stopping distance = reaction distance + braking distance The stopping distance is part of the danger zone.
Workspace	Area within which the robot may move. The workspace is derived from the individual axis ranges.
Automatic (AUT)	Operating mode for program execution. The robot moves at the programmed velocity.
User	The user of the medical device can be the management, employer or delegated person responsible for use of the medical device.
ESD	Electrostatic sensitive devices
Service life	The service life of a safety-relevant component begins at the time of delivery of the component to the customer. The service life is not affected by whether the safety-relevant component is used in a robot controller or elsewhere or not, as safety-relevant components are also subject to aging during storage.
Danger zone	The danger zone consists of the workspace and the stopping distances.
CRR	Controlled robot retraction CRR is an operating mode to which the system can be switched when the robot is stopped by the safety controller for one of the following reasons: • Robot violates an axis-specific or Cartesian monitoring space. • Orientation of a safety-oriented tool is outside the monitored range. • Robot violates a force or torque monitoring function. • A position sensor is not mastered or referenced. • A joint torque sensor is not referenced. Once the operating mode has been switched to CRR, the robot can be moved again. Note: This operating mode can only be selected in conjunction with the smartPAD.
KUKA smartPAD	see “smartPAD”
KUKA smartPAD-2	see “smartPAD”
Safety zone	The safety zone is situated outside the danger zone.

Safety stop	The safety stop is triggered by the safety controller, interrupts the work procedure and causes all robot motions to come to a standstill. The program data are retained in the case of a safety stop and the program can be resumed from the point of interruption. The safety stop can be executed as a Stop category 0, Stop category 1 or Stop category 1 (path-maintaining). Note: In this document, a safety stop of Stop category 0 is referred to as safety stop 0, a safety stop of Stop category 1 as safety stop 1 and a safety stop of Stop category 1 (path-maintaining) as safety stop 1 (path-maintaining).
smartPAD	Teach pendant The smartPAD is the teach pendant for the robot. It provides all the operator control and display functions required by the distributor and the system integrator (in the medical environment) for operating the robot during start-up, maintenance and diagnosis. 2 models exist: • smartPAD • smartPAD-2 In turn, for each model there are variants, e.g. with different lengths of connecting cables. The designation “KUKA smartPAD” or “smartPAD” refers to both models unless an explicit distinction is made.
Stop category 0	The drives are deactivated immediately and the brakes are applied.
Stop category 1	The robot is braked and deviates from the programmed path. The robot is brought to a standstill with the drives. As soon as an axis is at a standstill, the drive is switched off and the brake is applied. The internal electronic drive system of the robot performs safety-oriented monitoring of the braking process. Stop category 0 is executed in the event of a fault. Note: Stop category 1 is currently only supported by the LBR Med. For other manipulators, Stop category 0 is executed.
Stop category 1 (path-maintaining)	The robot is braked and stays on the programmed path. At standstill, the drives are deactivated and the brakes are applied. If Stop category 1 (path-maintaining) is triggered by the safety controller, the safety controller monitors the braking process. The brakes are applied and the drives are switched off after 1 s at the latest. Stop category 1 is executed in the event of a fault.
System integrator (plant integrator)	System integrators are people who safely integrate the LBR Med into a medical device and commission it.
T1	Test mode, Manual Reduced Velocity (≤ 250 mm/s) Note: With manual guidance in T1, the velocity is not reduced, but rather limited through a safety-oriented velocity monitoring in accordance with the safety configuration. Note: This operating mode can only be selected in conjunction with the smartPAD.

T2

Test mode, Manual High Velocity (> 250 mm/s permissible)

Note: This operating mode can only be selected in conjunction with the smartPAD.

3.2 Groups of persons



The operating and programming instructions for system integrators contain information for the medical device manufacturer and are not intended for the end user (e.g. medical personnel).

The following persons or groups of persons are defined for the LBR Med:

- System integrator / medical device manufacturer
- User
- Personnel



All persons working with the LBR Med must have read and understood the LBR Med documentation, including the safety chapter.

System integrator

The LBR Med must be safely integrated into the medical product by the system integrator.

The following list gives an indication as to the tasks for which the system integrator is responsible:

- Installing the LBR Med
- Connecting the LBR Med
- Conducting risk management for the medical product
- Use of the required safety equipment and safeguards
- Issuing the EC declaration of conformity
- Attaching the CE mark
- Creating the Instructions for Use for the medical product and the corresponding components for the overall system for the user and end user

User

The user must observe the labor laws and regulations. This includes e.g.:

- The user must comply with his monitoring obligations.
- The user must carry out briefing at defined intervals.
- The user must comply with the regulations relating to personal protective equipment (PPE).



The user must ensure that the robot controller is inaccessible to unauthorized personnel. The user is responsible for the storage and issuing of keys for the system (e.g. robot controller, smartPAD). The user must issue keys in accordance with the applicable laws, regulations and standards.

Personnel

Personnel must be instructed, before any work is commenced, in the type of work involved and what exactly it entails as well as any hazards which may exist. Instruction must be carried out regularly. Instruction is also required after particular incidents or technical modifications.

Personnel includes:

- System integrator
- Operators, subdivided into:
 - Start-up, maintenance and service personnel
 - Operating personnel
 - Cleaning personnel



Installation, exchange, adjustment, operation, maintenance and repair must be performed only as specified in the Instructions for Use for the relevant component of the LBR Med and only by personnel specially trained for this purpose.

Operators

The operator must meet the following preconditions:

- The operator must be trained for the work to be carried out.
- Work on the LBR Med must only be carried out by qualified personnel. These are people who, due to their specialist training, knowledge and experience, and their familiarization with the relevant standards, are able to assess the work to be carried out and detect any potential hazards.



Work on the electrical and mechanical equipment of the robot may only be carried out by KUKA Deutschland GmbH.

3.3 Workspace, safety zone and danger zone

Workspaces are to be restricted to the necessary minimum size in order to prevent danger to persons or the risk of material damage. Safe axis range limitations required for personnel protection are configurable.



Further information about the configuration of safely monitored axis limits is contained in the "Safety configuration" chapter of the operating and programming instructions for system integrators.

The danger zone consists of the workspace and the stopping distances of the robot. In the event of a stop, the robot is braked and comes to a stop within the danger zone. The safety zone is the area outside the danger zone.

The danger zone must be protected by means of physical safeguards, e.g. by light barriers, light curtains or safety fences. If there are no physical safeguards present, the requirements for collaborative operation must be examined and met where applicable in accordance with EN ISO 10218.

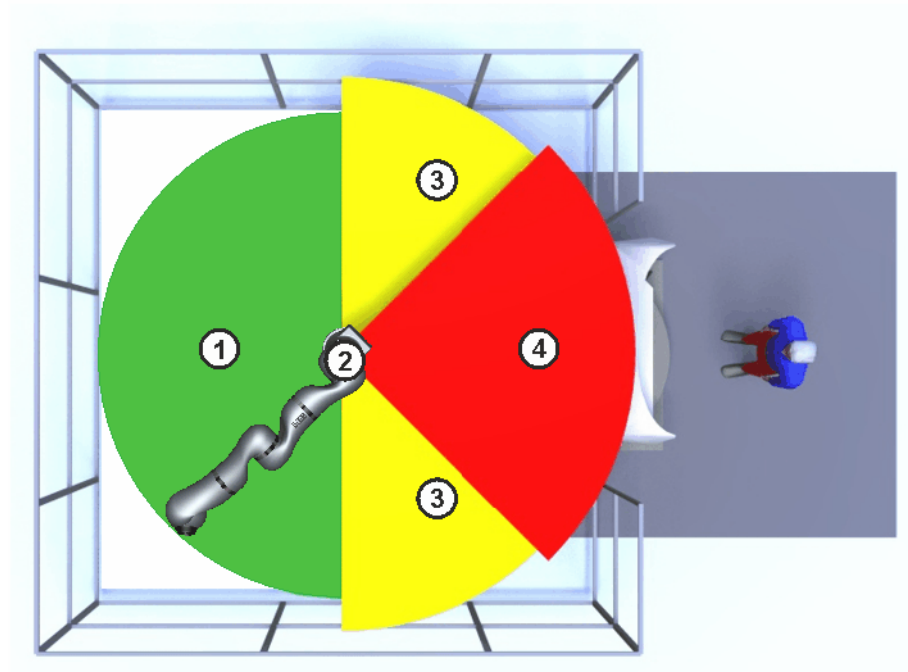


Fig. 3-1: Example: axis range A1

- | | | | |
|---|-------------|---|-------------------|
| 1 | Workspace | 3 | Stopping distance |
| 2 | Manipulator | 4 | Safety zone |

3.4 Triggers for safety-oriented stop reactions

Stop reactions are triggered in response to operator actions or as a reaction to monitoring functions and errors. The following tables show the different stop reactions according to the operating mode that has been set.

Overview

In Sunrise.OS a distinction is made between the following triggers:

- **Permanently defined triggers**
Permanently defined triggers for stop reactions and the associated stop category are preset by the system and cannot be changed. However, it is possible for the implemented stop reaction to be stepped up in the user-specific safety configuration.
- **User-specific triggers**
In addition to the permanently defined triggers, the user can also configure other triggers for stop reactions including the associated stop category.



Further information about configuring the safety functions is contained in the “Safety configuration” chapter of the operating and programming instructions of the system software for system integrators.

Permanently defined triggers

The following triggers for stop reactions are permanently defined:

Trigger	T1, T2, CRR	AUT
Operating mode changed during operation	Safety stop 1 (path-maintaining)	
Enabling switch released	Safety stop 1 (path-maintaining)	-
Enabling switch pressed fully down (panic position)	Safety stop 1 (path-maintaining)	-
Local E-STOP pressed	Safety stop 1 (path-maintaining)	
Error in safety controller	Safety stop 1	

User-specific triggers

When creating a new Sunrise project, the system automatically generates a project-specific safety configuration. This contains the following user-specific stop reaction triggers preconfigured by KUKA (in addition to the permanently defined triggers):

Trigger	T1, CRR	T2, AUT
Safety gate opened (operator safety)	-	Safety stop 1 (path-maintaining)
External E-STOP pressed	Safety stop 1 (path-maintaining)	
External safety stop	Safety stop 1 (path-maintaining)	



This default safety configuration is valid for the system software without additionally installed option packages or catalog elements. If additional option packages or catalog elements have been installed, the default safety configuration may be modified.

Triggers for manual guidance

If an enabling device is configured for manual guidance, the following additional triggers for stop reactions are permanently defined:

Trigger	T1, CRR	T2, AUT
Manual guidance enabling switch released	Safety stop 1 (path-maintaining)	-
Manual guidance enabling switch pressed fully down (panic position)	Safety stop 1 (path-maintaining)	-
Maximum permissible velocity exceeded while manual guidance enabling signal is set	Safety stop 1 (path-maintaining)	

A maximum permissible velocity of 250 mm/s is preconfigured for manual guidance. The maximum permissible velocity can be adapted.

The medical product manufacturer must evaluate the maximum permissible velocity in its risk management process and take the appropriate measures.

3.5 Safety functions

Safety functions are distinguished according to the safety requirements that they fulfill:

- Safety-oriented functions for the protection of personnel

The safety-oriented functions of the LBR Med meet the following safety requirements:

- **Category 3** and **Performance Level d** in accordance with EN ISO 13849-1
- **SIL 2** according to EN 62061

The requirements are only met on the following condition, however:

- All safety-relevant mechanical and electromechanical components of the LBR Med are tested for correct functioning during start-up and at least once every 12 months, unless otherwise determined in accordance with a workplace risk assessment. These include:
 - Local EMERGENCY STOP device on the teach pendant
 - Enabling device on the teach pendant
 - Enabling device on the hand guiding device (if present)
 - External enabling devices (if present)
 - Mode selector switch on the smartPAD (if used as teach pendant)
 - Safety-oriented outputs of the discrete safety interface



Details about safety parameters (e.g. PFH, SIL, Performance Level) are also available as a SISTEMA library. The library can be downloaded from the KUKA website.

- Non-safety-oriented functions

The non-safety-oriented functions of the LBR Med do not meet specific safety requirements.



DANGER

Risk of fatal injury due to non-operational safety functions or external safeguards

In the absence of operational safety functions or safeguards, the robot can cause death, severe injuries or damage to property.

- If safety functions or safeguards are dismantled or deactivated, do not operate the robot.



Integrate robot into safety system of medical device

During the integration of the LBR Med, the safety functions of the medical device must also be planned and designed. Death, severe injuries or damage to property may otherwise result.

- The LBR Med must be integrated into the safety system of the medical device.

3.5.1 Safety-oriented functions

The following safety-oriented functions are present and permanently defined in the LBR Med:

- EMERGENCY STOP device
- Enabling device

The following safety-oriented functions are preconfigured and can be integrated via the safety interface of the robot controller:

- Operator safety (= connection for the monitoring of physical safeguards)

- External EMERGENCY STOP device
- External safety stop 1 (path-maintaining)

Other safety-oriented functions may be configured, e.g.:

- External enabling device
- External safe operational stop
- Axis-specific workspace monitoring
- Cartesian workspace monitoring
- Cartesian protected space monitoring
- Velocity monitoring
- Standstill monitoring
- Axis torque monitoring
- Collision detection



Further information about configuring the safety functions is contained in the "Safety configuration" chapter of the operating and programming instructions of the system software for system integrators.

The preconfigured safety functions are described in the following sections on safety.

3.5.1.1 EMERGENCY STOP device

As standard, the EMERGENCY STOP device for the manipulator is the EMERGENCY STOP device on the smartPAD. The device must be pressed in the event of a hazardous situation or emergency.

Reaction of the LBR Med if the EMERGENCY STOP device is pressed:

- The robot stops with a safety stop 1 (path-maintaining).

Before operation can be resumed, the EMERGENCY STOP device must be turned to release it.



WARNING

Danger to life and limb due to tools and equipment without EMERGENCY STOP

If tools and other equipment connected to the robot are not integrated into the EMERGENCY STOP circuit, this can result in death, severe injuries or damage to property.

- Integrate tools and other equipment into the EMERGENCY STOP circuit if they could constitute a potential hazard.

If a holder is used for the teach pendant and conceals the EMERGENCY STOP device on the teach pendant, an external EMERGENCY STOP device must be installed that is accessible at all times.

If the smartPAD is unplugged, an external EMERGENCY STOP device must be installed that is accessible at all times.

(>>> [3.5.1.4 "External EMERGENCY STOP device" Page 32](#))

3.5.1.2 Enabling device

The enabling devices of the robot are the enabling switches on the smartPAD teach pendant as standard.

- **smartPAD:** 3 enabling switches
- **smartPAD-2:** 4 enabling switches

The enabling switches have 3 positions:

- Not pressed
- Center position
- Fully pressed (panic position)

In operating modes T1, T2 and CRR, the robot can only be moved if one of the enabling switches is held in the center position.

It is possible to hold several enabling switches in the center position simultaneously. This makes it possible to adjust grip from one enabling switch to another one.

In operating modes T1, T2 and CRR, the manipulator can be stopped in the following ways:

- Press at least one enabling switch down fully.
Fully pressing an enabling switch triggers a safety stop 1 (path-maintaining).
- Or release all enabling switches.
Releasing all (!) enabling switches held in the center position triggers a safety stop 1 (path-maintaining).



WARNING

Danger to life and limb due to lack of reaction when an enabling switch is released

Releasing one of multiple enabling switches held in the center position does not trigger a stop reaction.

If multiple switches are held in the center position, the robot controller cannot distinguish whether one of them was intentionally released or if it was unintentionally released as the result of an accident.

- Create awareness for the hazard.

If an enabling switch malfunctions (e.g. jams in the center position), the LBR Med can be stopped using the following methods:

- Press another enabling switch down fully.
- Actuate the EMERGENCY STOP device.
- Release the Start key.



WARNING

Danger to life and limb due to manipulation of enabling switches

The enabling switches must not be held down by adhesive tape or other means or tampered with in any other way. Death, severe injuries or damage to property may result.

- Carry out a visual inspection of the enabling switches.
- Rectify tampering or remove any foreign bodies.

3.5.1.3 “Operator safety” signal

The “operator safety” signal is used for monitoring physical safeguards, e.g. safety gates. In the default configuration, T2 and automatic operation are not possible without this signal. Alternatively, the requirements for collaborative operation must be examined and met where applicable in accordance with EN ISO 10218.

Reaction of the robot in the event of a loss of signal during T2 or automatic operation (default configuration):

- The robot stops with a safety stop 1 (path-maintaining).

By default, operator safety is not active in the modes T1 (Manual Reduced Velocity) and CRR, i.e. the signal is not evaluated.

**WARNING**

Following a loss of signal, automatic operation must not be resumed merely by closing the safeguard; the signal for operator safety must first be set by an additional device, e.g. by an acknowledge button. It is the responsibility of the system integrator to ensure this. This is to prevent automatic operation from being resumed inadvertently while there are still persons in the danger zone, e.g. due to the safety gate closing accidentally.

- This additional device must be designed in such a way that an actual check of the danger zone can be carried out first. Devices that do not allow this (e.g. because they are automatically triggered by closure of the safeguard) are not permitted.
- Failure to observe this may result in death to persons, severe injuries or considerable damage to property.

3.5.1.4 External **EMERGENCY STOP** device

Every operator station that can initiate a robot motion or other potentially hazardous situation must be equipped with an **EMERGENCY STOP** device. The system integrator is responsible for ensuring this.

Reaction of the LBR Med if the external **EMERGENCY STOP** device is pressed (default configuration):

- The robot stops with a safety stop 1 (path-maintaining).

External **EMERGENCY STOP** devices are connected via the safety interface of the robot controller. External **EMERGENCY STOP** devices are not included in the scope of supply of the LBR Med.

3.5.1.5 External **safety stop 1 (path-maintaining)**

The external safety stop 1 (path-maintaining) can be triggered via an input on the safety interface (default configuration). The state is maintained as long as the external signal is **FALSE**. If the external signal is **TRUE**, the robot can be moved again. No acknowledgement is required.

3.5.1.6 External **enabling device**

External enabling devices are required if it is necessary for more than one person to be in the danger zone of the LBR Med.

Multiple external enabling devices can be connected via the safety interface of the robot controller. External enabling devices are not included in the scope of supply of the LBR Med.

An external enabling device can be used for manual guidance of the robot. When enabling is active, the robot may only be moved at reduced velocity.

For manual guidance, safety-oriented velocity monitoring with a maximum permissible velocity of 250 mm/s is preconfigured. The maximum permissible velocity can be adapted.

The value for the maximum permissible velocity must be determined as part of a risk management process.

3.5.1.7 External **safe operational stop**

The safe operational stop is a standstill monitoring function. It does not stop the robot motion, but monitors whether the robot axes are stationary.

The safe operational stop can be triggered via an input on the safety interface. The state is maintained as long as the external signal is FALSE. If the external signal is TRUE, the robot can be moved again. No acknowledgement is required.

3.5.2 Non-safety-oriented functions

3.5.2.1 Mode selection

The LBR Med can be operated in the following modes:

- Manual Reduced Velocity (T1)
- Manual High Velocity (T2)
- Automatic (AUT)
- Controlled robot retraction (CRR)

Operating mode	Use	Velocities
T1	Programming, teaching and testing of programs.	• Program verification: reduced programmed velocity, maximum 250 mm/s • Jog mode: jog velocity, maximum 250 mm/s • Manual guidance: no limitation of the velocity, but safety-oriented velocity monitoring in accordance with the safety configuration
T2	Testing of programs	• Program verification: programmed velocity • Jog mode: not possible
AUT	Automatic execution of programs For robots with and without a higher-level controller	• Program mode: programmed velocity • Jog mode: not possible
CRR	CRR is an operating mode to which the system can be switched when the robot is stopped by the safety controller for one of the following reasons: • Robot violates an axis-specific or Cartesian monitoring space. • Orientation of a safety-oriented tool is outside the monitored range. • Robot violates a force or torque monitoring function. • A position sensor is not mastered or referenced. • An axis torque sensor is not referenced. Once the operating mode has been switched to CRR, the robot can be moved again.	• Program verification: reduced programmed velocity, maximum 250 mm/s • Jog mode: jog velocity, maximum 250 mm/s • Manual guidance: no limitation of the velocity, but safety-oriented velocity monitoring in accordance with the safety configuration

Mode selector switch

The user can change the operating mode via the connection manager. The connection manager is a view that is called by means of the mode selector switch on the smartPAD.

The mode selector switch for LBR Med:

- **With key**
It is only possible to change operating mode if the key is inserted.

3.5.2.2 Velocity monitoring in T1

The reduced velocity in T1 does not constitute a safety-rated reduced speed in the standard safety configuration, i.e. the maximum permissible velocity of 250 mm/s in T1 is not subjected to safety-oriented monitoring.

If the application requires safety-oriented velocity monitoring in T1, this can be added in the safety configuration. The safety option KUKA Sunrise.SafeOperation provides the monitoring function *Cartesian velocity monitoring* for this purpose.



Further information about configuring safety-oriented velocity monitoring for T1 is contained in the “Safety configuration” chapter of the operating and programming instructions of the system software for system integrators.

3.5.2.3 Software limit switches

The axis ranges of all axes on the robot are limited by means of non-safety-oriented software limit switches. These software limit switches only serve as protection for the robot and are preset in such a way that the robot is stopped under servo control if the axis limit is exceeded, thereby preventing damage to the mechanical system.

3.6 Service phases of the robot system

The following service phases have been defined for the robot system:

- **Transportation**
- **Start-up and recommissioning**
- **Normal operation**
- **Malfunction**
- **Cleaning**
- **Maintenance**
- **Repair**
- **Decommissioning**
- **Storage**
- **Disposal**

Full responsibility for the service phases indicated lies with the medical product manufacturer.

The medical product manufacturer must provide the end customer the necessary information for operation of the medical product (overall system) during the service phases of normal operation, malfunction, cleaning and maintenance.

3.7 Additional protective equipment

3.7.1 Jog mode

In the operating modes T1 (Manual Reduced Velocity), T2 (Manual High Velocity) and CRR, the robot controller can only execute programs in jog mode. This means: It is necessary to hold down an enabling switch and the start key in order to execute a program.

- Releasing the enabling switch on the smartPAD triggers a safety stop 1 (path-maintaining).
- Pressing the enabling switch on the smartPAD fully down triggers a safety stop 1 (path-maintaining).
- Releasing the Start key triggers a stop of Stop category 1 (path-maintaining).

3.7.2 Labeling on the LBR Med

All plates, labels, symbols and marks constitute safety-relevant parts of the LBR Med.



CAUTION

If the plates and labels are removed for integration into a medical product, the medical product manufacturer must evaluate this in its risk management process and take the necessary measures.

Labeling on the LBR Med consists of:

- Identification plates
- Warning labels
- Safety symbols
- Designation labels
- Cable markings
- Rating plates



Further information is contained in the technical data of the Instructions for Use for the components of the LBR Med.

3.7.3 External safeguards

The medical product manufacturer must evaluate the selection of external safeguards in its risk management process and take the necessary measures for the medical product / overall system.

Access of persons to the danger zone of the LBR Med must be prevented by means of safeguards. Alternatively, the requirements for collaborative operation must be examined and met where applicable in accordance with EN ISO 10218. It is the responsibility of the system integrator to ensure this.

Physical safeguards must meet the following requirements:

- They must be examined according to requirements and meet EN ISO 14120 where applicable.
- They prevent access of persons to the danger zone and cannot be easily circumvented.

- They are sufficiently fastened and can withstand all forces that are likely to occur in the course of operation, whether from inside or outside the enclosure.
- They do not, themselves, represent a hazard or potential hazard.
- The prescribed minimum clearance from the danger zone is maintained.

Safety gates (maintenance gates) must meet the following requirements:

- They are reduced to an absolute minimum.
- The interlocks (e.g. safety gate switches) are linked to the configured operator safety inputs of the robot controller.
- Switching devices, switches and the type of switching conform to the requirements of Performance Level d and category 3 according to EN ISO 13849-1.
- Depending on the hazard situation: the safety gate is additionally safeguarded by means of a locking mechanism that only allows the gate to be opened if the robot is safely at a standstill.
- The device for setting the signal for safety, e.g. the button for acknowledging the safety gate, is located outside the space limited by the safeguards.



Further information is contained in the corresponding standards and regulations. These also include EN ISO 14120.

Other safety equipment

Other safety equipment must be integrated into the system in accordance with the corresponding standards and regulations.

3.8 Safety measures

3.8.1 General safety measures

The LBR Med may only be used in perfect technical condition in accordance with its intended use and only by safety-conscious persons. Operator errors can result in personal injury and damage to property.

It is important to be prepared for possible movements of the robot even after the robot controller has been switched off and locked. Incorrect installation (e.g. overload) or mechanical defects (e.g. brake defect) can cause the robot to sag. If work is to be carried out on a switched-off LBR Med, the robot must first be moved into a position in which it is unable to move on its own, whether the payload is mounted or not. If this is not possible, the robot must be secured by appropriate means.



DANGER

Risk of fatal injury due to non-operational safety functions or external safeguards

In the absence of operational safety functions or safeguards, the robot can cause death, severe injuries or damage to property.

- If safety functions or safeguards are dismantled or deactivated, do not operate the robot.